1. Problem

1.1. Describe the problem from a general perspective.

Academic performance of students is a common indicator of the quality of education and student success. Student performance is usually measured in grades, GPA, CGPA, exam results, or completion of courses. However, a review of literature shows that the grade of students are affected by more than one influence rather than one single cause.

The most common factors across reviews and empirical studies fall into a few broad groups: prior academic performance, attendance in classroom, family and socioeconomic background, learning environment.

The key problem is the difficulty of identifying students at risk of academic failure or dropout at an early stage. This can negatively affect student retention, graduation rates, and overall educational outcomes, while also creating broader social impacts as education plays a critical role in individual development and societal progress. Therefore, effective analytical and predictive approaches are needed to identify at-risk students early and enable timely interventions to minimise these potential impacts.

1.2. Highlight the business/research need.

The need of the hour for institutions is to improve student outcomes, reduce failure and drop-out, better allocate participation, and strengthen their institutional performance indicators such as retention, ranking, graduate employability etc. Many studies clearly claim that early prediction assists universities in identifying low performing students and intervening prior to final failure.

The need for research is just as strong because academic performance is complex, nonlinear and zone specific and missed in manual analysis. As noted in the reviews, no model or variable has been found to universally explain performance across all settings, with results depending on the dataset, educational level, and features.

As per the reviews, there exists no consistent model or single variable that accounts for performance in all situations, as results depend upon the dataset, level of education and type of features included.

1.3. Define the overall project objective.

The objective of this work is to study student academic performance as a multi-dimensional educational issue impacted by academic, behavioral, demographic, family, institutional factors, and utilize relevant datasets to study which factors are highly associated with student success or risk of underperformance.

1.4. Briefly indicate how individual datasets contribute to the problem.

Ngoc An Nguyen has shared a dataset that helps identify whether students' academic history variables contribute to their predicted performance.

Quang Trung Nguyen provided a dataset in descriptive statistics that contains behavioral indicators of a student.

Akil Venkateshkumar's data set provides a good number of demographic and socioeconomic variables to test the pretest logic.

2. Literature Review

Costa et al. (2024) and Kocsis and Molnar (2025) analysed two systematic reviews. The main aim of these studies was to assess the combined results of academic success, including GPA and ECTS credits. The results suggested that educational success depends on both layered and multidimensional determinants of academic achievement. These included academic engagement and class attendance. For instance, "the richer the family background, the greater the impact of financial conditions on the student's results" (Costa et al., 2024). However, this family variable had no further impact beyond the upper layer. The same was true of previous performance and bonus effects.

Namoun and Alshanqiti (2021) reviewed Batool et al. (2023) and found that machine learning applications in education can be used to predict the quality and quantity of learning. Both Namoun and Alshanqiti (2021) and Batool et al. (2023) agree that subject and pedagogical experts should be encouraged to create better models for predicting student learning. Behavioral and academic characteristics are some of the main predictors of student learning (Namoun and Alshanqiti, 2021). Prior academic performance (e.g., prior CGPA) and active engagement behaviours (e.g., classroom attendance and digital learning behaviour) are consistent predictors of academic success (Batool et al., 2023).

El Jihaoui et al. (2025) established various predictive patterns with empirical support from several studies. According to El Jihaoui et al. (2025), Vadivel et al. (2023), among many other studies, academic performance was the "best" predictor of success, based on a data set with over 1 million students. The study found that academic performance explained 76.6 percent of the variance in "success," while SES accounted for a far smaller but still substantial 6.02%. According to (2023), if a family has a low income and the parents provide little or no support, this can lead to a reduction in the child's academic performance, as the child does not attend school regularly and joins the labour market at an early age.

3. Approach

3.1. Describe the overall analytical approach/framework.

The analytical framework follows five steps: dataset description and acquisition, data preparation, research question formulation, EDA and statistical analysis, and presentation of key insights using figures or tables. Python tools are used for data analysis and visualisation.

3.2. Outline the techniques and models planned for Stage 2.

Using the exploratory analysis from Stage 1, Stage 2 will predict the final grade of the students through the numerous factors present in the individual datasets contributed by the group members. Model development processes will use data pre-processing, feature selection, and train-test splitting analysis. Across datasets within the group, three regression models will be applied consistently: Linear regression, Decision Tree, and Random Forest. Assessment of model performance will be based on R^2, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE). The performance of the model with the best overall prediction will be compared across the individual datasets.

4. Individual Analysis and Findings

4.1. Student Name: Ngoc An Nguyen

Student ID: 26507564

4.1.1. Dataset Selection and Research Question

This individual analysis is based on the student performance dataset, which includes 395 records and 33 attributes relating to students' academic achievement, learning habits, family backgrounds and social characteristics. The final grade (G3) will serve as the dependent variable. Important attributes include previous grades (G1, G2), previous academic failures, hours spent studying, number of absences, education levels of parents, and behavioral traits.

Link: [Student-performance]

Because it contains multiple academic and non-academic variables that can be used for comparisons, the student performance dataset is an appropriate resource for studying differences in student academic achievement.

Research Question: What are the primary academic, behavioral, family and background factors associated with students' final academic achievement (G3)?

4.1.2. Data Preparation

Data preparation occurred through the use of Python with a focus on Pandas, NumPy, SciPy and Matplotlib. Data quality checks were conducted for missing data, duplicate observations, inconsistent categorical variables and invalid numeric ranges. All 395 observations remained after data quality checks due to the absence of missing data, duplicate observations or invalid grade values.

Potential outliers were examined via the interquartile range (IQR) method; however, outliers were not automatically excluded when representing valid observations. Subsequent analyses involved descriptive statistics and visualizations to assess distributions and relationships among variables. Descriptive statistics and exploratory analysis were emphasized throughout the course.

4.1.3. Exploratory Data Analysis and Key Insights

Prior academic achievement demonstrated the strongest association with final grade. G2 was positively correlated with G3 at a very high level ($r = 0.905$, $p < .001$). G1 was also positively correlated with G3; although at a somewhat lower level ($r = 0.801$, $p < .001$). These correlations indicate that students who performed well in prior courses typically received high final grades.

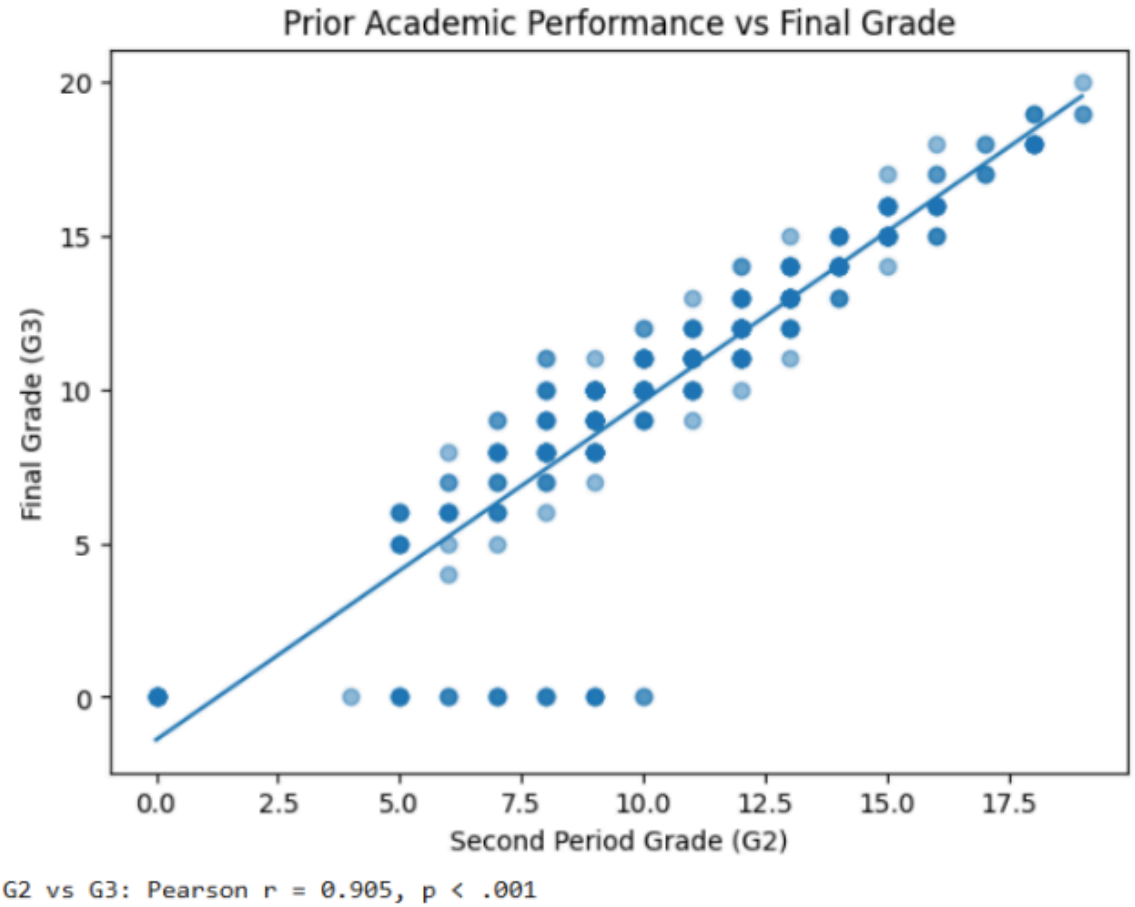


Figure 1. Relationship between Second Period Grade (G2) and Final Grade (G3).

Other factors were less strongly correlated with final achievement. Prior academic failure exhibited the strongest negative correlation with final achievement (Spearman rho = -0.361, $p < .001$). Students without previous academic failures had an average G3 of 11.25, while students with three previous academic failures had an average G3 of 5.69.

Behavioral factors were significantly less correlated with final achievement: Going out demonstrated a relatively weak negative correlation with final achievement (rho = -0.166), while hours spent studying demonstrated a relatively weak positive correlation with final achievement (rho = 0.105).

Family factors exhibited relatively weak correlations with final achievement. Maternal education (Medu) was positively correlated with G3 (rho = 0.225, $p < .001$), while paternal

education (Fedu) was positively correlated with G3; although at a slightly lower level (rho = 0.170, p < .001).

Among categorical variables, students who planned to attend post-secondary institutions had an average G3 of 10.61, while students who did not plan to attend post-secondary institutions had an average G3 of 6.80. However, comparisons based on these two groups must be viewed cautiously because the two groups were substantially different in size (375 vs. 20 students).

Performance_Group	Count	Mean_G3	Mean_G1	Mean_G2	Mean_Failures	Mean_Studytime	Mean_Absences	Mean_Medu
Low	130	5.38	7.76	6.82	0.69	1.95	6.76	2.57
Medium	165	11.22	10.95	11.04	0.23	2.00	6.05	2.61
High	100	15.62	14.93	15.23	0.04	2.21	3.78	3.21

Table 1. Key academic and behavioural characteristics by final-grade performance group.

4.1.4. Conclusion

Overall, prior academic achievement — specifically G2 — was most strongly correlated with final grade, followed by previous academic failures. Behavioral and family factors were less strongly correlated with final grade. These correlations are exploratory relationships and cannot be considered as causal relationships.

4.2. Student Name: Quang Trung Nguyen

Student ID: 26643559

4.2.1. Dataset Description

The accessible dataset on students' behavioural and academic engagement is available at Kaggle. The dataset has 1000 student records. These records are basically an entity with numerical variables: study time, attendance, hours of sleep, previous day's grade, and final exam score. Also, it has categorical variables: internet access, extracurricular activity participation, part-time job, and final grade.

This analysis focuses on students' behavioural activities and classroom engagement; therefore, factors outside these areas are excluded from the analysis. The selected variables include study hours, attendance, sleep hour, internet access, extracurricular activities, part-time job and final exam score. The final_grade variable is excluded because it is derived from final exam score and therefore does not provide additional independent information for analysis.

4.2.2. How the data was acquired

The dataset was obtained from Kaggle where it is published publicly by the dataset author. The exact source and method were not documented in any specific way. As a result, we are unable to verify the data collected. The dataset is available on Kaggle: [Student Performance and Study Habits Dataset]

4.2.3. Individual research question

How do behavioural factors, particularly study time, attendance, and sleep hours, relate to students' final grades?

How are engagement and access-related factors, including internet access, extracurricular activities, and part-time employment, associated with students' final grades?

4.2.4. Exploratory Data Analysis

There is a relatively strong moderate positive linear correlation ($r = 0.57$) between study time and final exam scores. An increase of 0.322 in R^2 is attributed to study hour increase with a score increase of 3.97. Attendance has a weak association ($r = 0.26$), reflecting a 0.29-point increase in score per 1% increase in attendance ($R^2 = 0.069$). There is little correlation ($r = 0.15$) with sleep, where one-hour more of sleep is associated with 1.27 more points ($R^2 = 0.022$). There is a statistically significant association for all three relations ($p < 0.05$).

The correlation heatmap shows the three predictors are weakly correlated, which indicates they have little linear association. All in all, a relationship and regression outcome shows that study time had the strongest association with academic performance which was then followed by attendance and sleep.

	Variable	Slope	R-squared	p-value
0	study_time_hours	3.969	0.322	0.0
1	attendance_percent	0.292	0.069	0.0
2	sleep_hours	1.270	0.022	0.0

Table 2. Linear Regression Results for Predictors of Final Exam Score

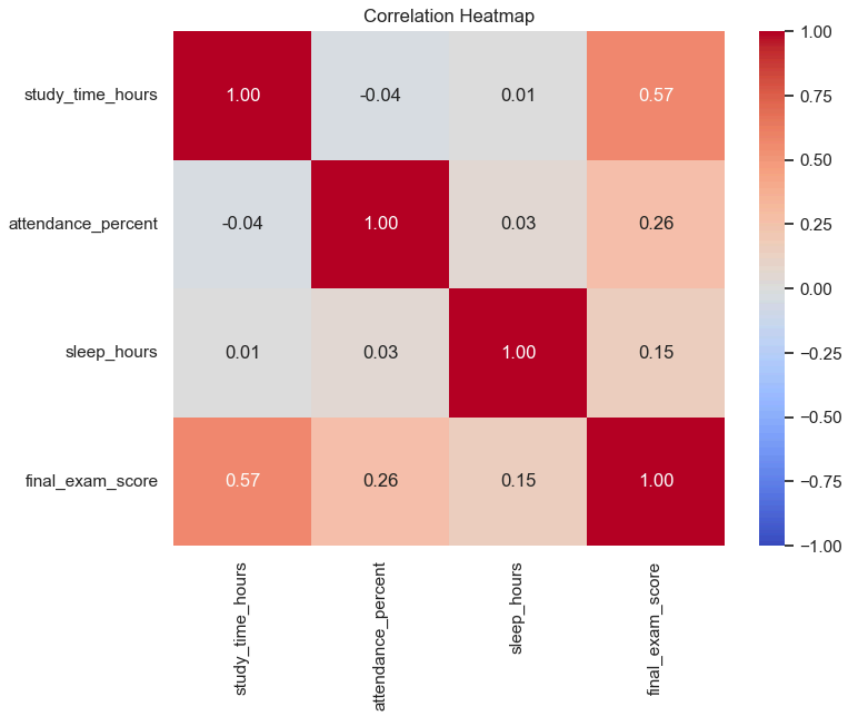


Figure 2. Correlation Matrix of Study Factors and Final Exam Score

For categorical variables, internet access is associated with higher exam scores. Students with internet access scored an average of 84.28, compared with 79.26 without access, a difference of 5.02 points, which was statistically significant ($p < 0.001$). Part-time work is associated with lower scores: students with a job averaged 81.30 compared with 84.58 for those without one, a significant difference of 3.29 points, which was also significant ($p < 0.001$). In contrast, extracurricular activities showed no significant difference, with only a 0.45-point mean difference ($p = 0.488 > 0.001$). Overall, mean differences indicate the direction and size of group effects, while p-values determine whether these differences are statistically significant.

	Variable	Yes Mean	No Mean	Mean Difference	t-statistic	p-value	Significant
0	internet_access	84.276	79.258	5.018	5.216	0.000	Yes
1	extracurricular_activities	83.737	83.284	0.453	0.694	0.488	No
2	part_time_job	81.295	84.582	-3.287	-4.555	0.000	Yes

Table 3. Independent Samples t-Test Results for Categorical Variables

4.2.5. Key Insights

Among numeric factors, study time had the strongest positive association with final exam scores, followed by attendance, while sleep duration showed a weak positive association. Besides, Internet access was significantly associated with higher scores, whereas part-time employment was associated with lower scores. Finally, extracurricular activities showed no significant association with exam performance.

4.2.6. Data preparation

The dataset was checked for missing values, duplicate records, inconsistent categories, invalid values, and potential outliers. No missing or duplicate records were identified, and categorical values were consistent. Numerical variables were checked against their expected ranges. student_id was excluded as an identifier, while the remaining variables were retained for analysis.

4.2.7. Tools used

Python, with pandas for data preprocessing, matplotlib and seaborn for data visualization, and scipy.stats for statistical analysis, including ttest_ind for comparing the means of two groups and linregress for linear regression analysis.

4.3. Student Name: Akil Venkateshkumar

Student ID: 26244259

4.3.1. Dataset source, size, and features

The dataset utilized for this analysis pertains to the demographic and socioeconomic aspects of student performance in exams, which is publicly accessible on Kaggle. It includes 1,000 student records with eight attributes: gender, race/ethnicity, parental education level, lunch classification (standard or free/reduced, serving as an indicator for household income), participation in a test preparation course, and three continuous outcome measures — math, reading, and writing scores. In contrast to the academic-history dataset utilized by Ngoc An Nguyen, which records previous grades and evaluations, this dataset focuses on isolating background and equity-related factors, enabling the project to examine if demographic and family situations are linked to performance without being influenced by prior academic records.

4.3.2. How the data was acquired

Methods of data collectionThe CSV file was downloaded directly from Kaggle as the dataset. It is artificially created yet based on authentic distributions found in educational surveys, which mitigates privacy issues while maintaining realistic connections between demographic factors and results. No external API or web scraping was needed; the file was directly imported into a pandas DataFrame in Google Colab for analysis.

Link dataset: [Students performance in exams]

4.3.4. Research question

How significantly are demographic and family-background factors — such as gender, parental education, and socioeconomic indicator (type of lunch) — related to variations in student academic achievement, and do these influences remain consistent across the three subject scores?

4.3.5. Exploratory data analysis and key insights

EDA mean scores (average of math, reading, and writing) were calculated according to parental education level, lunch type, and gender. Scores rose gradually with higher parental education levels, ranging from 63.1 (high school) to 73.6 (master's degree) — an ANOVA test confirmed this as statistically significant ($F=10.75$, $p<0.001$). Students with standard lunch averaged a score of 70.8, whereas those on free or reduced lunch had an average of 62.2, leading to a statistically significant difference of 8.6 points according to the t-test ($t=-9.58$, $p<0.001$). Gender indicated a significant difference: female students averaged 69.6, while male students averaged 65.8 ($t=4.17$, $p<0.001$), although this gap was not as marked as the influences of parental education or lunch type. Together, these results support the idea that parental education, lunch type, and gender should be regarded as important demographic and socioeconomic factors influencing academic success, consistent with the SES literature addressed in Section 2. (see Figures 4–6.)

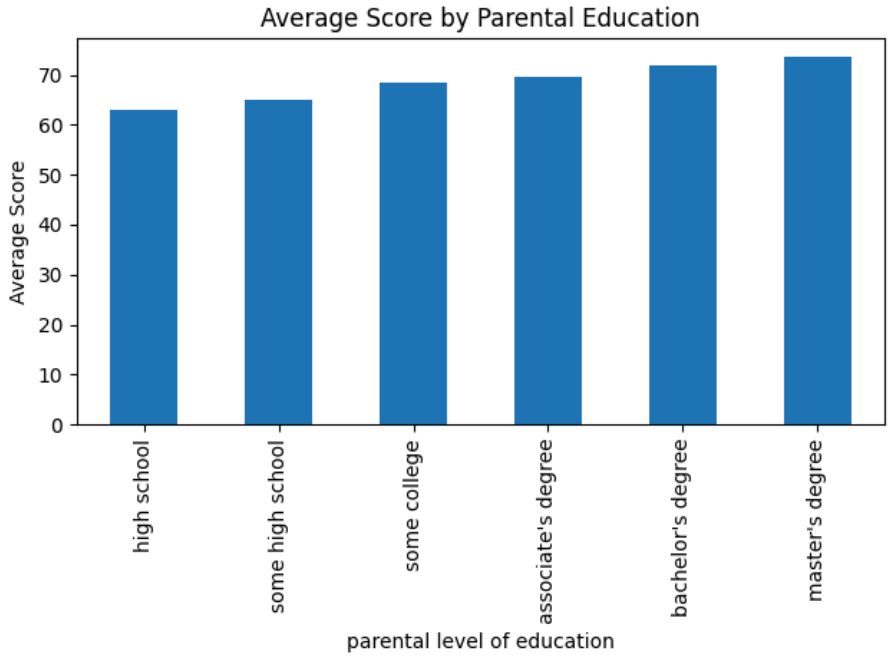


Figure 4: Average score by parental education level

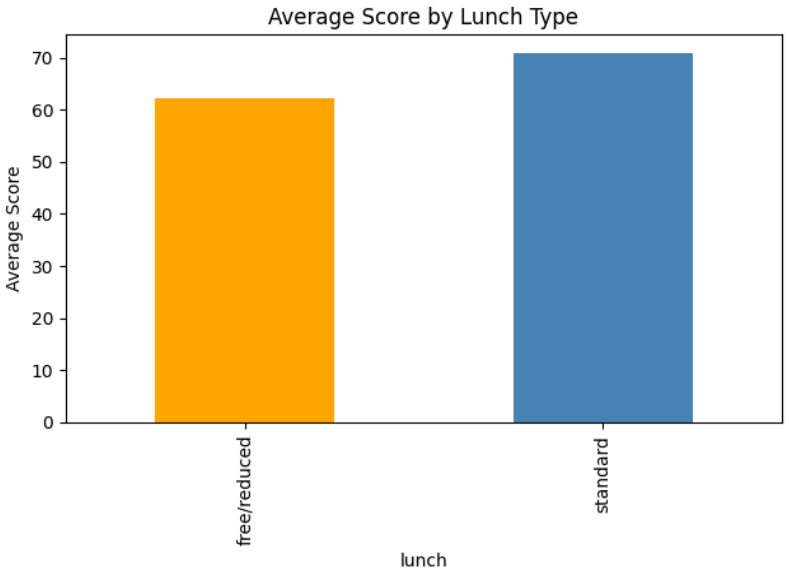


Figure 5: Average score by lunch type

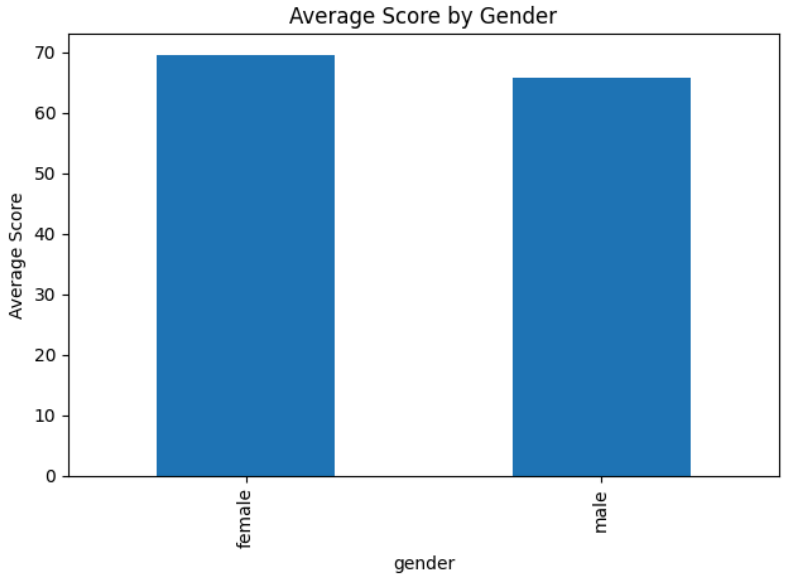


Figure 6: Average score by gender

4.3.6. Data preparation

The dataset needed little cleaning since there are no missing values in any of the eight columns. Categorical variables (gender, race/ethnicity, parental education, lunch, test preparation) were kept in their initial form for group comparison, and the three subject scores were combined into a single composite performance score to facilitate group comparisons, while individual scores remain accessible for subject-level analysis if required in Stage 2.

4.3.7. Tools used

Python (pandas for data handling, matplotlib for visualization, scipy for statistical testing) in Google Colab.

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